Exercise 5.4

1. Construct the following angles using protractor and draw a bisector to each of the angle using ruler and compass.
   (a) 60° (b) 100° (c) 90° (d) 48° (e) 110°

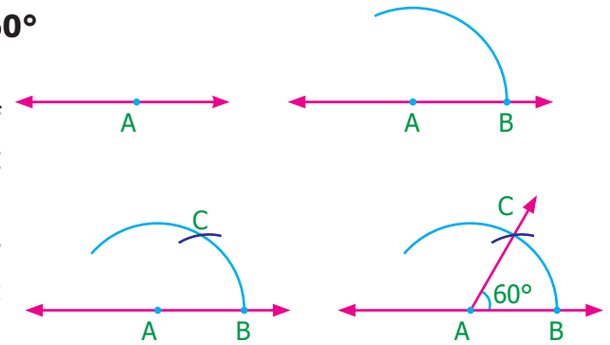
5.4.3 Construction of special angles without using protractor.

(i) Construction of angle of measure 60°

Step 1: Draw a line. Mark a point A on it.

Step 2: With A as center draw an arc of convenient radius to the line to meet at a point B.

Step 3: With the same radius and B as center draw an arc to cut the previous arc at C.



Step 4: Join AC. Then $\angle BAC$ is the required angle with the measure 60°.

(ii) Construction of angle of measure 120°.

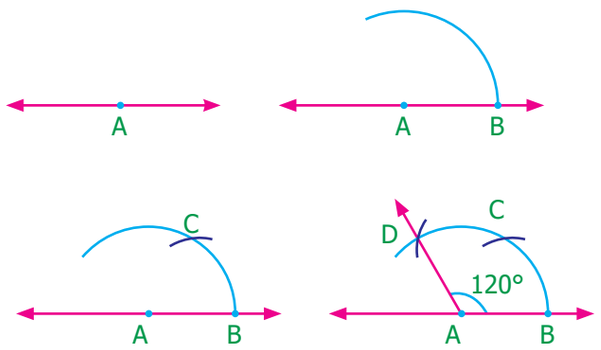
We know that there are two 60° agnels in 120°. Hence, to construct 120°, we can construct two 60° angles consecutively as follows.

Step 1: Draw a line. Mark a point A on it.

Step 2: With A as center, draw an arc of convenient radius to the line at a point B.

Step 3: With the same radius and B as center, draw an arc to cut the previous arc at C.

Step 4: With the same radius and C as center, draw an arc to cut the arc drawn in step 2 at D.



Step 5: Join AD. Then $\angle BAD$ is the required angle with measure 120°.

(iii) Construction of angle of measure 30°

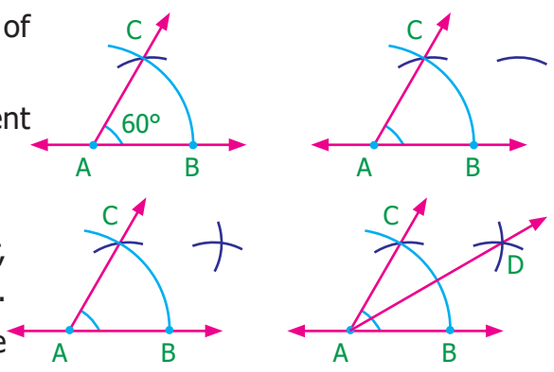
Since 30° is half of 60°, we can construct 30° by bisecting the angle 60°.

Step 1: Construct angle 60° [Refer Construction of angle of measure 60° (i)].

Step 2: With B as center, draw an arc of convenient radius in the interior of $\angle BAC$.

Step 3: With the same radius and C as center, draw an arc to cut the previous arc at D.

Step 4: Join AD. Then $\angle BAD$ is the required angle with measure 30° [Think about $\angle DAC$?]



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